ML LAB EXPT – 3

Machine Learning lab(BCSL606): Dr. S. V. Manjaragi

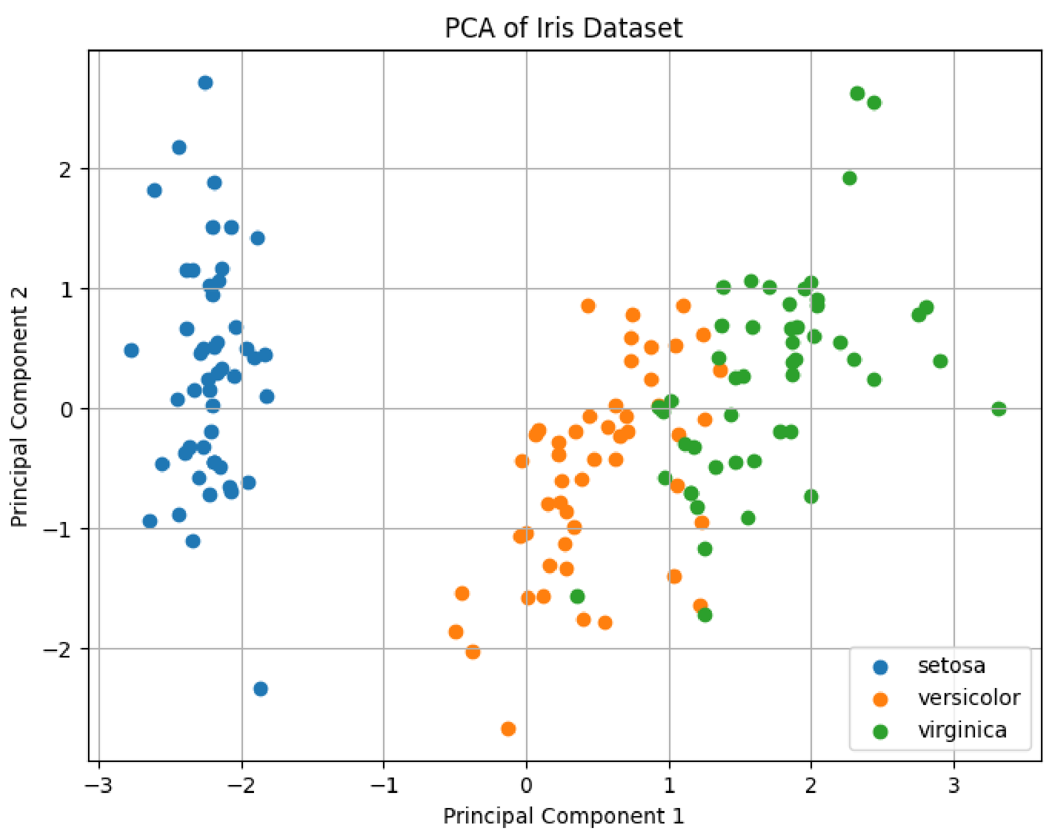
Aim of the Experiment:

Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.

```
# Assuming you have the iris dataset downloaded as 'iris.csv' in your
current directory

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load the Iris dataset from the CSV file
try:
    df = pd.read_csv('iris.csv')
except FileNotFoundError:
    print("Error: iris.csv not found.")
    exit()

# Separate features and target variable
X = df.drop('species', axis=1)
y = df['species']

# Standardize the features (mean=0, variance=1) each feature has a mean of
0 and a variance of 1.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply PCA with 2 components
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Create a DataFrame for the PCA results
pca_df = pd.DataFrame(data=X_pca, columns=['PC1', 'PC2'])
pca_df['species'] = y

# Print explained variance ratio
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

```python
# Visualize the results
plt.figure(figsize=(8, 6))
for species in pca_df['species'].unique():
    subset = pca_df[pca_df['species'] == species]
    plt.scatter(subset['PC1'], subset['PC2'], label=species)

plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA of Iris Dataset')
plt.legend()
plt.grid(True)
plt.show()
```

Output:



Machine Learning lab(BCSL606): Dr. S. V. Manjaragi